HYDRALIS

Software Requirements Specification (SRS)

Flood Early Warning, Dispatch and Safe Evacuation Platform

Professional specification for the Hydralis mobile application and dispatcher web application, balancing hackathon delivery with post-hackathon commercial feasibility.

Document version	1.0
Date	11 April 2026
Prepared for	Hackathon concept development and post-hackathon product planning
Scope	Cross-platform citizen mobile application, dispatcher web application, integrations and supporting operational requirements

Positioning statement. Hydralis is defined as a last-mile flood warning and evacuation orchestration product that complements official alerting channels rather than replacing them. The initial commercial focus is local authorities, utilities and critical-site operators that need practical decision support, citizen guidance and multi-channel message export at a lower cost and with lower deployment complexity than large national-scale platforms.

Document map

This specification combines product scope, market positioning and detailed software requirements for the Hydralis mobile and dispatcher applications.

1. Introduction and purpose

This Software Requirements Specification defines the Hydralis product vision, commercial positioning, software scope and detailed requirements for two coordinated applications: a cross-platform citizen mobile application and a dispatcher web application. The specification is intentionally optimized for a hackathon MVP that can be demonstrated quickly and for a post-hackathon product that remains commercially realistic for public-sector and critical-infrastructure buyers.

1.1 Product scope

Hydralis is a flood early warning and evacuation orchestration platform. The citizen mobile application is designed to receive high-urgency warnings, present clear instructions, guide users toward safe locations and provide citizen status feedback. The dispatcher web application is designed to ingest hazard inputs, manage incidents, approve and publish alerts, maintain a live operational map and export broadcast-ready messages for external channels.

1.2 Product boundaries

- Hydralis complements official public alerting mechanisms and emergency procedures; it does not replace national cell-broadcast infrastructure or the legal authority required to issue official alerts.
- The product focuses on flood and flash-flood scenarios in the initial release in order to keep the MVP clear, operationally credible and commercially focused.
- The initial offering is software-first, with optional low-cost sensor integration rather than a hardware-heavy deployment model.
- Radio and television support is implemented through standards-based outputs and media-ready message packages, not through channel-specific playout automation in the MVP.

1.3 Stakeholders and user classes

Stakeholder	Primary goal
Emergency coordinator	Approves severe alerts, monitors response quality and ensures policy compliance.

Stakeholder	Primary goal
Dispatcher / operator	Uses the web console to review triggers, validate incidents, publish guidance and track field status.
Citizen / resident	Receives alerts, follows route guidance, checks in as safe or requests help, and submits situational reports.
Municipal administrator	Configures safe locations, templates, user roles, map layers and tenant-level settings.
Field technician	Installs or maintains gateways and sensors and monitors their health status.
Partner broadcaster or agency	Receives CAP output, concise radio/TV text or export packages for downstream dissemination.

2. Market feasibility and competitive positioning

The central business premise behind Hydralis is that many authorities and site operators do not need a national command platform; they need an affordable operational layer that translates hazard inputs into local decisions, targeted alerts and practical citizen guidance. Commercial feasibility improves significantly when the product is positioned as a complement to existing official alert systems, not a direct replacement.

Commercial thesis. Hydralis is most feasible as a B2G and B2B2C product sold to local authorities, utilities and critical-site operators that need flood-specific operational guidance, low-complexity deployment and standards-based downstream outputs.

2.1 Priority customer segments

Segment	Operational need	Priority
Municipalities and small-to-mid-size cities	Need affordable incident workflows, local routing and citizen communication without deploying a large national-scale platform.	Primary
County-level emergency and water-management bodies	Need a shared map, local telemetry, alert review and consistent message export across multiple communities.	Primary
Utilities and critical infrastructure operators	Need to protect staff, assets and service continuity in sites exposed to flooding or access disruption.	Secondary
Industrial parks, logistics campuses and large facilities	Need site-level alerting, assembly point guidance and status visibility for employees or contractors.	Secondary

Segment	Operational need	Priority
Broadcasters and partner agencies	Need clean, short, standardized outputs, but are more likely to act as partners than as direct primary buyers.	Partner

2.2 Competitive landscape and product positioning

Category	Typical strength	Gap Hydralis addresses
Official public warning systems	High authority and wide reach for public warning.	Usually weaker as local operational workspaces for routing, shelter management and community-specific decision support.
Enterprise emergency communication suites	Strong multi-channel communication and mature governance features.	Often expensive, broad and operationally heavy for smaller flood-prone municipalities that need a narrower, simpler product.
Flood intelligence and prediction tools	Strong modelling and hazard visualization.	Often stop at awareness; they may not provide citizen-facing guidance, local approval workflows or practical dispatch operations.

2.3 Differentiation principles

- Flood-specific rather than generic multi-hazard in the initial release.
- Local last-mile execution: geofenced guidance, safe-location management and citizen check-in.
- Low deployment friction: software-first, optional sensor layer, standards-based outputs.
- Operator-friendly workflow for smaller teams that do not have dedicated GIS or emergency software specialists.
- Commercial compatibility with official systems by exporting CAP and media-ready messages instead of trying to displace existing alert channels.

2.4 Indicative business model

Package	Included value	Typical buyer
Starter	Dispatcher console, citizen app, templates, CAP export and safe-location management.	Municipal pilot or single-site deployment
Operations Plus	Adds sensor integration, reporting, multilingual templates and richer audit features.	County-level or utility deployment
Enterprise / Critical Site	Adds SLA, integration work, role customization and multi-site operations.	Infrastructure operator or large campus

3. System overview

Hydralis is designed as a coordinated software system composed of a citizen mobile client, a dispatcher web client, a central backend platform and an integration layer for hazard data, mapping and local telemetry. The architecture below shows the product boundary and the main information flow.

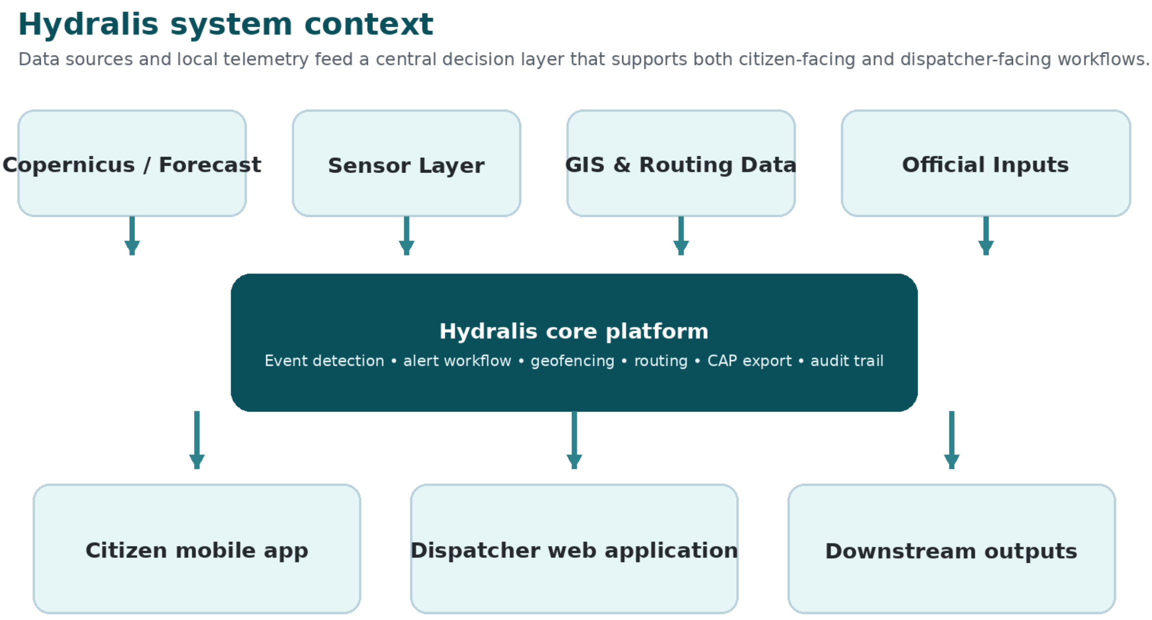

Figure 1. High-level Hydralis system context.

3.1 Core solution components

- Citizen mobile application built for Android and iOS through a shared cross-platform codebase.
- Dispatcher web application for operators, coordinators and administrators.
- Backend services for event detection, alert workflow, geofencing, routing and audit logging.
- Data integration layer for forecast feeds, map layers, safe locations and sensor telemetry.
- Output layer for push alerts, CAP payloads, short broadcaster scripts and structured exports.

3.2 Primary operational workflows

- Forecast-driven workflow: a forecast or imported hazard signal triggers an incident suggestion, which is reviewed and published by an authorized dispatcher.
- Sensor-driven workflow: local telemetry exceeds configured thresholds and raises an operator-visible alert with location and device health context.
- Citizen guidance workflow: the user receives a critical alert, sees the impacted zone, receives voice and visual instructions and follows the safest route to a designated safe location.
- Broadcast workflow: the same validated incident generates CAP output and concise scripts for radio, television or partner systems.

4. Assumptions, dependencies and constraints

- The first release assumes at least one reliable upstream flood or weather data source and at least one maintained map source for roads and safe locations.
- Availability of Galileo-capable device positioning depends on handset chipset and operating-system exposure of GNSS capabilities.
- Safe-route quality depends on the accuracy of road network data, road-closure updates and safe-location configuration maintained by the tenant.

- Some official data sources may require formal agreements, licensing approval or API onboarding before production use.
- The MVP may simulate selected integrations where real-time production access is not available during the hackathon.
- The initial software scope excludes direct control of telecom cell broadcast infrastructure and direct automation of every broadcaster distribution chain.
- Where connectivity is poor, the mobile application must degrade gracefully by using cached instructions and the most recent known incident context.

5. Platform-wide functional requirements

The following requirements apply to the Hydralis platform as a whole and are shared across the mobile and dispatcher applications where relevant. Priority values use a MoSCoW model: Must, Should and Could.

5.1 Event detection and incident creation

ID	Requirement	Priority
FR-CORE-001	The system shall ingest at least one forecast-based hazard input and at least one manual incident input.	Must
FR-CORE-002	The system shall support configurable rules for incident suggestion based on threshold breaches, severity bands or operator-defined trigger logic.	Must
FR-CORE-003	The system shall create an incident object with geospatial extent, severity, timestamps, source attribution and workflow state.	Must
FR-CORE-004	The system shall allow operators to edit, confirm, reject or merge automatically suggested incidents.	Must
FR-CORE-005	The system shall preserve an audit trail of source data, rule evaluation and operator actions for each incident.	Must

5.2 Alert workflow, geofencing and outputs

ID	Requirement	Priority
FR-CORE-006	The system shall support alert lifecycle states including draft, review, approved, published, updated and closed.	Must
FR-CORE-007	The system shall support geofenced targeting by polygon, administrative area or selected safe-location catchment.	Must

ID	Requirement	Priority
FR-CORE-008	The system shall generate a standards-based CAP output for approved alerts.	Must
FR-CORE-009	The system shall generate concise broadcaster-ready message variants from the same approved alert content.	Should
FR-CORE-010	The system shall support multilingual alert templates and variable substitution for location, severity and instruction text.	Must
FR-CORE-011	The system shall allow follow-up updates and all-clear messages to be linked to the originating incident.	Must

5.3 Reporting and administration

ID	Requirement	Priority
FR-CORE-012	The system shall log operator actions, publication history and user-visible message versions.	Must
FR-CORE-013	The system shall provide incident summary exports in human-readable and machine-readable formats.	Should
FR-CORE-014	The system shall support tenant-level configuration for templates, safe locations, thresholds and access control.	Must
FR-CORE-015	The system shall preserve configuration history for critical operational settings.	Should

6. Mobile application requirements

The Hydralis mobile application is the citizen-facing component. It must deliver very clear alerts, calm but urgent instructions and practical guidance during a stressful event. The application shall prioritize clarity, accessibility, low cognitive load and graceful offline behavior.

6.1 Alerting and incident communication

ID	Requirement	Priority
FR-MOB-001	The mobile application shall receive high-priority push alerts and display them as interruptive critical notifications when permitted by the operating system.	Must
FR-MOB-002	The mobile application shall support a high-urgency alert mode with distinctive sound, vibration and optional spoken warning text.	Must
FR-MOB-003	The mobile application shall display alert severity, affected area, time, clear instructions and recommended next action on the primary alert screen.	Must
FR-MOB-004	The mobile application shall distinguish between warning, escalation, route update and all-clear message types.	Must

6.2 Map, routing and safe locations

ID	Requirement	Priority
FR-MOB-005	The mobile application shall display the user’s current location, the relevant incident zone and one or more recommended safe destinations on a map.	Must
FR-MOB-006	The mobile application shall provide route guidance toward a configured safe location and refresh the recommendation when the incident geometry or route constraints change.	Must
FR-MOB-007	The mobile application shall present a safe-location card containing destination type, address, distance, status and human-readable instructions.	Must
FR-MOB-008	The mobile application shall cache the latest incident map, destination and instruction set for use in degraded connectivity conditions.	Must

6.3 Citizen status, reporting and resilience

ID	Requirement	Priority
FR-MOB-009	The mobile application shall allow a user to mark status as safe, need help or cannot evacuate.	Must
FR-MOB-010	The mobile application shall allow the user to submit a geotagged field report with optional photo or short note, subject to connectivity and consent.	Should
FR-MOB-011	The mobile application shall support a low-bandwidth mode that prioritizes text instructions over map tiles and media.	Should
FR-MOB-012	The mobile application shall maintain a local history of recent alerts and status submissions visible to the user.	Should

6.4 Accessibility, language and privacy

ID	Requirement	Priority
FR-MOB-013	The mobile application shall support Romanian and English content in the initial release and a localization framework for additional languages.	Must
FR-MOB-014	The mobile application shall support accessibility features including scalable text, high contrast, screen-reader compatibility and optional voice guidance.	Must
FR-MOB-015	The mobile application shall support a guest usage mode for alert reception, with optional account-based features layered on top.	Should
FR-MOB-016	The mobile application shall provide user-facing privacy controls for location sharing, report submission and retention of personal status information.	Must

7. Dispatcher web application requirements

The Hydralis dispatcher application is the operational control surface for authorities and other approved organizations. It must support fast situational understanding, low-friction incident publishing and transparent auditability.

7.1 Authentication, authorization and dashboard

ID	Requirement	Priority
FR-DSP-001	The dispatcher application shall require authenticated access and shall support role-based permissions.	Must
FR-DSP-002	The dispatcher application shall present a live map view with layers for incidents, sensors, safe locations, reports and route constraints.	Must
FR-DSP-003	The dispatcher application shall provide summary panels for active incidents, pending approvals, sensor issues and user status updates.	Must
FR-DSP-004	The dispatcher application shall allow operators to filter the workspace by geography, severity, source and workflow state.	Must

7.2 Incident management and decision support

ID	Requirement	Priority
FR-DSP-005	The dispatcher application shall allow an operator to create, edit, duplicate, update and close incidents.	Must
FR-DSP-006	The dispatcher application shall support map-based geofence editing and preview of the affected population or area.	Must
FR-DSP-007	The dispatcher application shall support templated message creation with manual editing before publication.	Must
FR-DSP-008	The dispatcher application shall provide rule-based recommendations for severity, instructions and destination selection when upstream data is available.	Should

ID	Requirement	Priority
FR-DSP-009	The dispatcher application shall display source provenance, threshold breach context and related events to support operator judgment.	Must

7.3 Population guidance, shelters and broadcast outputs

ID	Requirement	Priority
FR-DSP-010	The dispatcher application shall maintain a registry of safe locations with type, capacity, accessibility notes and operational status.	Must
FR-DSP-011	The dispatcher application shall allow operators to assign or recommend destinations by incident, geography or user segment.	Must
FR-DSP-012	The dispatcher application shall generate CAP output, concise radio/TV message text and text suitable for text-to-speech generation.	Must
FR-DSP-013	The dispatcher application shall visualize citizen check-ins, help requests and field reports on the incident map.	Must

7.4 Device operations, reporting and administration

ID	Requirement	Priority
FR-DSP-014	The dispatcher application shall allow registration and health monitoring of connected sensor devices or gateways.	Should
FR-DSP-015	The dispatcher application shall record operator comments, handover notes and timeline events within each incident.	Must
FR-DSP-016	The dispatcher application shall export incident summaries, timeline reports and configuration snapshots for after-action review.	Should
FR-DSP-017	The dispatcher application shall allow authorized administrators to manage users, templates, language packs and routing settings.	Must

ID	Requirement	Priority
FR-DSP-018	The dispatcher application shall provide a clear separation between operational views and configuration views.	Should

8. External interface requirements

8.1 Upstream data and telemetry interfaces

Interface	Format / mode	Minimum requirement
Forecast / EO feed	REST, file import or scheduled ingestion	The system shall support ingestion of forecast or hazard data with timestamps, severity and geospatial context.
Sensor telemetry	MQTT, HTTP or gateway API	The system shall support ingestion of device telemetry including location, reading value, timestamp and health state.
Map and routing data	GeoJSON, vector tiles or routing API	The system shall support road network, polygon and point layers needed for incident zones and safe-location guidance.
Administrative inputs	Web form or API	The system shall support manual overrides, closures, destination status and authority-authored instructions.

8.2 Downstream outputs and partner interfaces

Output	Representation	Minimum requirement
Mobile push channel	Critical and standard notifications	The system shall deliver approved alerts to registered mobile devices through a supported push provider.
CAP export	Structured XML payload	The system shall output approved alerts in CAP-compliant structure for downstream systems.
Media pack	Short text and TTS-ready script	The system shall generate concise message variants suitable for broadcaster handoff.
Operational export	CSV, PDF or JSON	The system shall export incident logs, status summaries and reports for partners or after-action review.

8.3 Mobile device capabilities

Capability	Requirement
GNSS location	The mobile application shall use device location services and shall leverage Galileo-capable positioning where available through the operating system.
Audio and vibration	The mobile application shall use device speaker and vibration channels for high-urgency warning presentation, subject to platform permissions.
Camera and media attachment	The mobile application may use camera or gallery access for optional citizen field reports.
Local storage	The mobile application shall use secure local storage for cached alerts, recent maps and user preferences needed for degraded connectivity.

9. Non-functional requirements

ID	Category	Requirement	Priority
NFR-001	Availability	The dispatcher application should target at least 99.5% monthly service availability excluding planned maintenance.	Should
NFR-002	Performance	The dispatcher incident dashboard shall load the default view in under 3 seconds under normal operating conditions.	Must
NFR-003	Alert latency	An approved alert shall be handed off to the mobile notification provider within 10 seconds of publication.	Must
NFR-004	Scalability	The system architecture shall support tenant growth through horizontal scaling of stateless application services.	Should
NFR-005	Security	All client-server communication shall be encrypted in transit and sensitive stored data shall be protected at rest.	Must

ID	Category	Requirement	Priority
NFR-006	Privacy	The system shall minimize personal data collection and provide configuration support for GDPR-aligned retention and deletion policies.	Must
NFR-007	Accessibility	The web application should align with WCAG 2.2 AA and the mobile application shall support accessible text, contrast and assistive technologies.	Must
NFR-008	Interoperability	The system shall use standards-based APIs and common data formats such as REST/JSON, GeoJSON, CSV and CAP where applicable.	Must
NFR-009	Resilience	The mobile application shall operate in degraded mode with cached instructions when the network is unstable.	Must
NFR-010	Auditability	Critical operator actions and published alert versions shall be recorded in immutable audit logs.	Must
NFR-011	Maintainability	Configuration for templates, thresholds and destination data shall be externalized so that routine updates do not require app redeployment.	Should
NFR-012	Observability	The backend shall expose logs, metrics and health checks sufficient to diagnose ingestion failures, publish failures and device issues.	Should
NFR-013	Localization	The system shall support at least Romanian and English in the initial release and a localization process for future languages.	Must
NFR-014	Battery and data efficiency	The mobile application should minimize background battery drain and avoid unnecessary large downloads during active incidents.	Should

10. MVP scope, release plan and acceptance criteria

To remain both hackathon-ready and commercially credible, the first demonstrable release must prove the value of the operational loop rather than the breadth of every future capability. The MVP therefore centers on one strong flood scenario from trigger to citizen guidance.

10.1 Hackathon MVP scope

Scope type	Description
Included in MVP	Dispatcher map with incident creation, geofence editing, message publishing and safe-location registry.
Included in MVP	Citizen mobile experience with critical alert, voice prompt, map guidance and safe check-in.
Included in MVP	At least one upstream hazard or forecast input plus a simulated or real local sensor signal.
Included in MVP	CAP export plus concise radio/TV message pack generated from the same approved alert.
Included in MVP	Incident timeline and basic audit history.
Deferred after MVP	Large-scale sensor fleet operations, advanced occupancy integration and family account workflows.
Deferred after MVP	Direct broadcaster system integrations, telecom-grade public warning integrations and multi-hazard expansion.

10.2 Acceptance criteria for the first demonstrable release

- A dispatcher can create or confirm a flood incident, define the affected zone and publish an alert in less than two minutes during a scripted demo.
- A mobile user receives the alert, hears or sees clear instructions and can open a route toward a configured safe location.
- A safe check-in or help request submitted from the mobile device appears in the dispatcher timeline and map view.
- The same incident can produce both a mobile alert and a CAP or media-ready export without requiring duplicated authoring.
- A forecast or telemetry signal can generate an operator-visible event suggestion with enough context for review.

10.3 Post-hackathon release priorities

- Strengthen tenant administration, identity integration and SLA-grade monitoring.
- Formalize data-source agreements and production routing data quality controls.
- Expand safe-location capacity management, richer citizen status analytics and deployment tooling for sensor packages.

11. Risks and mitigations

Risk	Impact	Mitigation
Data access or licensing delays	Could block production-grade forecasting or official data integration.	Design the platform with source abstraction and demo-ready simulated feeds; pursue formal onboarding in parallel.
Trying to replace official warning channels	Would create regulatory and go-to-market friction.	Position Hydralis as a local operational complement and standards-based exporter.
Connectivity loss during incidents	Could weaken citizen guidance and dispatcher visibility.	Cache recent instructions locally and support degraded-mode workflows.
Alert fatigue or false positives	Could reduce trust and adoption.	Use approval workflows, clear severity rules and post-incident review data.
Hardware rollout complexity	Could slow sales cycles and increase support burden.	Keep hardware optional and start software-first with a minimal telemetry layer.
Long public-sector procurement cycles	Could delay revenue after the hackathon.	Pilot with one municipality or critical-site operator and package clear ROI around response coordination and compliance.

Appendix A. Glossary and abbreviations

Term	Definition
CAP	Common Alerting Protocol; a structured standard for exchanging emergency alerts.
Dispatcher	An operator or coordinator using the Hydralis web application to manage incidents and publish guidance.
EO	Earth observation data used to enrich hazard awareness and mapping.
Geofence	A spatial boundary used to target an alert, workflow or routing rule.
GNSS	Global Navigation Satellite System; Hydralis can leverage Galileo-capable device positioning where available.
Incident	A Hydralis event record representing a hazard situation, its area, state and associated messages.
MVP	Minimum Viable Product; the smallest release that proves the operational value of Hydralis.

Term	Definition
Safe location	A designated shelter, assembly point or destination recommended during an incident.
TTS	Text-to-speech output used for spoken warning or broadcaster handoff.